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1. Executive Summary

This report evaluates, verifies, corrects, completes, and then attempts to solve the “Active-Set Bridge Conjecture” formulation deposited in *external audits/unifying object/deepseek formulation.txt* (316 lines: the conjecture with assumptions A1–A5, a five-step proof sketch, three testable consequences, a status paragraph, and the model's own five-point self-critique with a revised statement). The evaluation uses three evidence classes: the mathematical structure of multi-parametric linear programming; the executed M1/M3/M3b numerical record on the deterministic lexicographic-pFBA engine (iML1515); and two new measurements, M4a and M4b, designed and executed for this report to attack the formulation's own declared missing ingredient — the scaling limit A4. The verdict is mixed in a precise way: the formulation's structural half (piecewise-affinity, event concentration, active-set substrate) is correct and already machine-verified, while its central analytic claim — an ϵ^2 -scaled holonomy limit equal to the geometric curvature — is false in both layers in which it could be interpreted, and it is false for reasons that are now theorems, not opinions.

The repaired formulation delivered here (Section 5) replaces the conjecture with a five-part architecture: a static curvature-measure theorem (S), an intrinsic defect theorem with a validated unfolding transport (G), a new structural lemma discovered during verification (N1, “no loose kinks”), an atomicity obstruction theorem that explains exactly why the strong form cannot be proven (N), and a first-order dynamic commutator law (D) that M4a measured to slope 0.998. What survives of the original bridge is substantial: the active-set skeleton is the common carrier of every object, exactly as the six unifying-object audits converged in the joint assessment. What is abandoned is the infinitesimal ϵ^2 pointwise identification, and its abandonment is now a proven necessity rather than a stylistic caution.

slope 1.00, not 2

M4a measured the dynamic commutator scaling on 76 gene pairs at six perturbation depths: 64 pairs are exactly commutative ($\chi \equiv 0$) and the 9 nonzero pairs scale linearly (median slope 0.998, exact halving ratios across five octaves). The ϵ^2 law of A4 is falsified in the only layer where it was meaningful.

1e-14 / 1e-11

The corrected unfolding transport equals the corner-angle defect to machine precision on synthetic piecewise-affine maps (including a 190.6° cone), and composes to the exact identity on real iML1515 interfaces away from codim-2 strata (5/5 flat controls). The geometric machinery the conjecture needed exists; it was attached to the wrong limit.

Two findings go beyond the audit task. First, the atomicity obstruction (Theorem N) converts the six-audit consensus that the strong-form unification “is not provable” into a localized reason: for a fixed model the curvature of the flux graph is supported on the codim-1 and codim-2 skeletons, so any ϵ^2 -normalized loop limit is either zero (generic points) or divergent (skeleton points), and the only sound limits are mesoscopic (defect density, an assumption on model families) or dynamic (Theorem D). Second, the edge-crossing census of the (glucose, oxygen) plane shows the skeleton is operationally dense precisely where the second-derivative mass concentrates — up to nine boundary crossings inside a single grid cell of the overflow-metabolism fan, versus exactly two in flat regions — which is the empirical face of the same obstruction.

2. The Formulation Under Evaluation

The file advances a single conjecture with supporting apparatus. Its setting is a genome-scale metabolic network solved as a parametric FBA family with a fixed lexicographic pFBA tie-break, making the optimal flux map $v(\theta)$ unique for almost every θ and piecewise affine on a finite polyhedral decomposition of the parameter space. On the chamber complex the file defines “affine-extension transition maps” T_{ij} between the affine representatives of adjacent regions, asserts the cocycle condition on triple overlaps, and assembles a “discrete holonomy” H_γ as the product of transitions along a closed piecewise-smooth loop γ .

Five assumptions carry the statement. A1 (nondegenerate FBA) grants uniqueness and a basis-like active set off a measure-zero set. A2 (polyhedral regularity) makes the active-set boundaries transverse hypersurfaces of a finite complex. A3 (geometric embedding) posits a smooth embedding Φ of the parameter space into a Riemannian manifold B carrying a vector bundle, a connection ∇ with curvature Ω^∇ , and a viability field V , from which the geometric viability-weighted curvature κ_{geom} is defined as the V -contraction of Ω^∇ . A4 (scaling limit) asserts that the ε^2 -normalized loop holonomy converges to $\Omega^\nabla(\Phi_*X, \Phi_*Y)$ and its norm to κ_{geom} , independently of the loop shape within a fixed tangent plane. A5 (flatness on strata) declares the discrete connection flat on each open stratum. The conjecture is then the displayed limit; the proof sketch runs piecewise affinity, jump encoding, second-order holonomy, Dirac-supported D^2v , and a comparison step; and three consequences are claimed — second-order concentration on active-set events, double-knockout non-additivity as a rectangle holonomy, and the time-course object as integrated curvature. The status paragraph concedes that only the scaling limit A4 is missing.

The file's second half is a self-critique proposing five fixes: replace the infinite-dimensional affine-map holonomy by a finite-dimensional tangent-space transport P_{ij} ; construct the connection explicitly from KKT basis exchanges (rank-one updates); restrict the limit to regular crossing points or restate it distributionally; make the parallel transport precise as an orthogonal projection; and replace the holonomy estimate by an exact second-order expansion $H_\varepsilon - I = \Delta_{e1}\Delta_{e2}\varepsilon^2 + O(\varepsilon^3)$. A revised compact statement incorporates these. Our audit confirms the instincts behind all five fixes while finding specific technical errors in three of them (Section 4), and it finds the claimed empirical support overstated in one place (C5) — the file asserts that M1 and M3 “confirm the first two consequences to numerical precision,” which is true for consequence 1 and contradicted for consequence 2.

3. Verification Against the Executed Record

Every checkable claim of the formulation was re-verified against the M1/M3/M3b artifacts (commit lineage 2fe264b, f9f8634, 6cdd563) rather than against the summaries. The record is consistent with the formulation's structural claims: the 12 parametric sweeps of M1 measured second-order response mass on material active-set events at 0.99999996 (glucose sweep), 0.99999946 (oxygen sweep), and 0.934–1.0 across the eight gene knockdowns, with piecewise-affine segment residuals at or below 8×10^{-14} relative and two negative controls behaving as a flat map must (the unused-pathway knockdown `kd_aceA` produced zero events and $D2 \equiv 0$; the `iJO1366` glucose replication stayed within a single critical region with $D2$ at the 10^{-11} noise floor). The 2,779 double knockouts of M3 support the substrate claim: rerouting epistasis tracks active-set footprint overlap with Spearman $\rho = 0.865$ on the dR mask ($p \approx 0$), all 40 synthetic lethals are isozyme redundancies, and growth epistasis anticorrelates with flux epistasis at $\rho = -0.705$.

Formulation element	Verdict	Evidence
$v(\theta)$ piecewise affine on a finite complex (A1/A2)	Verified	M1: affine residuals $\leq 8e-14$ relative on 12 sweeps; M4b Jacobians exact within chambers (affine-consistency test)
v continuous across interfaces	Verified	M1 D1 kinks, not jumps; M4b continuity diagnostics consistent with kink slopes; no tears at analyzed loci
D^2v a measure on active-set interfaces	Verified	M1 D2 mass 0.934–1.0; M4b 1D cut through a codim-2 region: 100.0000000% on 11 events
Consequence 1 (second-order concentration)	Confirmed	M1 headline result, both models, two controls
Consequence 2: ϵ_{ij} = rectangle holonomy	Contradicted	M3b: χ vs $ \epsilon $ Spearman -0.347 full panel ($p = 6.8e-6$), -0.07 within non-SL ($p = 0.43$): the optimum non-additivity and the transient non-commutativity are independent
Consequence 3: κ'' = curvature measure	Imprecise	Missing smooth term $2\ v\ ^2$; pointwise equality should be an integral identity (Theorem S(iv))
The status-paragraph confirmation claim (M1 and M3 “confirm consequences 1–2 to numerical precision”)	Overstated	True for consequence 1 only (C5)
Cocycle of T_{ij} on overlaps	True, and fatal	A globally consistent affine representative system forces $H_\gamma = I$ exactly (C1/C6)
A4 ϵ^2 scaling limit	Falsified	M4a: slope 0.998 on the nonzero stratum; Theorem N for the static layer

Table 1. Formulation claims against the executed record.

One verification deserves emphasis because it changes the interpretation of M3b. The closed-loop non-return measured in M3b (66.25% of genotype loops return to the wild-type genotype but not to the initial state, median displacement 109 L1 units) was described there as a holonomy. The release-identity check run in M4a (6/6 pairs, bit-exact) shows that every relaxation projection is the identity — the state after a loop fails to return because the greedy L1-MOMA dynamics never climbs back once displaced, not because a connection rotates the state around a plaquette. The M3b loop statistic is irreversibility; the genuine non-commutativity is the open-path commutator χ , and it is χ that M4a scales. This distinction propagates into the repaired Theorem D.

4. Mathematical Audit

4.1 C1 — The central object is trivially flat

The affine-extension holonomy H_γ is exactly the identity for every closed loop, because v is a single-valued continuous function. The chamber representatives of a common region agree on shared interfaces (this is the file's own cocycle claim, verified), so composing transition maps around any loop telescopes back to the starting representative. The conjecture's central display therefore evaluates to $0 = \Omega^V$ for the static map. This is not a normalization defect that a better limit could repair: single-valuedness forbids any holonomy on the function's own representatives. The nontrivial static curvature is the distributional Jacobian-jump measure on codim-1 interfaces — the object M1 measured — together with the defect measure on codim-2 strata (Theorem G). Any correct bridge must carry these two measures, not a loop limit of affine maps.

4.2 C2 — The ϵ^2 law fails in both layers

In the static layer the law cannot even be posed (C1). In the dynamic layer — the sequential L1-MOMA adjustment, the only path-dependent object in the executed program — M4a measured the commutator $\chi(\epsilon)$ under a flux-relative scaled knockdown family in which $\epsilon = 1$ is the M3 full knockout and $\epsilon = 0$ is the wild-type operating point. The measured law is $\chi(\epsilon) = c \cdot \epsilon + O(\epsilon^2)$ on the non-decoupled stratum (Section 6). The mechanism is structural: each knockout adds an $O(1)$ face to the binding set and thereby changes the tangent cone of the feasible set by an $O(1)$ jump, and the second adjustment step inherits a different cone depending on the order, so the order mismatch accumulates at first order. The ϵ^2 commutator regime would require the two adjustment maps to share a common linear part at the operating point; the data exclude this at every sampled scale. A4 is thus resolved negatively in the only place it was meaningful, and the honest replacement is Theorem D's first-order law — which is, notably, a new falsifiable prediction for graded perturbation experiments.

4.3 C3 — The proposed transport is not a connection

The self-critique's finite-dimensional repair transports tangent vectors by orthogonal projection from F_i to F_j . Projections are not invertible — $P_{ji}P_{ij} \neq I$ whenever the subspaces differ — so the transition system violates the groupoid inverse axiom, holonomy is ill-defined, and back-tracking does not cancel. The correct object is the minimal-rotation (unfolding, polar) map: the rotation about the shared edge that carries one face's tangent plane onto the other's. It is an invertible isometry, fixes the common edge exactly, reduces to the identity in the coplanar limit, and makes curvature concentrate on the codim-2 skeleton. This report validated that transport in two independent ways: on synthetic piecewise-affine maps, where the loop composition equals the corner-angle defect to 10^{-14} including a 190.6° cone, and on real iML1515 interfaces, where crossing one interface twice composes to the identity at 10^{-11} with axis and angle constant along the interface (Section 7).

4.4 C4 — A3 is unfalsifiable as stated

A3 merely assumes the existence of an embedding Φ , a connection, and a viability field. Under that assumption nothing can be proven or refuted — any map embeds into any geometry trivially (a constant embedding already forces $\Omega = 0$), and the conjecture's right-hand side is unconstrained. The repair is constructive: take the graph map $G(\theta) = (\theta, v(\theta))$ with the induced Euclidean metric as the Riemannian object, and define κ_{geom} as the viability-weighted discrete Gauss curvature (angle defect) of this graph. The embedding, the metric, and the connection are then fixed by the FBA data itself, the static bridge becomes a theorem (Theorem G), and the only remaining identification — with the manuscript's continuum objects — is coarse-grained and empirical, which is exactly what the E28 item already tracks. This also matches the joint assessment's Layer-1 adjudication: opus's mixed difference as the combinatorial definition, with the geometric thread organized smoothly rather than assumed existentially.

4.5 C5 — The claimed confirmations are overstated

The file's status paragraph asserts that M1 and M3 confirm the first two consequences “to numerical precision.” The record supports half of that sentence. Consequence 1 is confirmed outright. Consequence 2's central identity — that the epistasis ϵ_{ij} equals, in the small-perturbation limit, the holonomy of the perturbation rectangle — is contradicted: M3b measured both statistics on the same 160-pair panel and found no positive association (full-panel Spearman -0.347 driven by the synthetic-lethal structure; within non-lethal pairs -0.07 , $p = 0.43$). What M3 actually established is different and stronger in its own way: non-additivity aligns with

active-set footprint overlap ($\rho = 0.865$). The repaired reading keeps both facts: the optimum-level non-additivity and the transient-level non-commutativity are distinct signatures carried by the same skeleton, and the bridge must transport them separately rather than identify them.

4.6 C6 and the additional corrections

C6: A5 (flatness on strata) is not merely compatible with the conjecture — it is the reason the conjecture's conclusion cannot follow. A globally consistent representative system is exactly a flat connection; only transports on objects that are not single-valued (tangent frames across kinked folds, dynamic states under greedy adjustment) can carry holonomy, and then only as $O(1)$ defects at codim-2 strata. Additional corrections: A1 is achievable but non-trivial — the engine's own construction history is the existence proof, since vertex degeneracy in the parsimony stage (sum|v| ties to 13 digits with 0.69-magnitude flux flips between warm starts) had to be broken by a seeded third-stage tie-break; on measure-zero strata the deterministic selection may tear, which the analysis must tolerate (none were found at the analyzed loci). The rank-one update claim is correct within a stage under nondegeneracy, and the three-stage lexicographic structure makes the connection a tower — matching the factored curvature datum of the joint assessment's Layer 1. The proof-sketch step “holonomy = integral of the second-order measure” is replaced by the true structure equations: the codim-1 jump measure and the codim-2 defect measure are linked by the unfolding composition (the incident dihedral kinks determine the vertex defect).

5. The Repaired Formulation (Active-Set Bridge v2)

Setting: a compact polyhedral parameter space Θ environmental bounds and/or genetic capacities, the optimal flux map v of the genome-scale LP under the fixed three-stage lexicographic tie-break, its chamber complex C with codim-1 skeleton $C^{(p-1)}$ and codim-2 skeleton $C^{(p-2)}$. The full statement, with proofs and the status table, is preserved as the standalone document download/Active_Set_Bridge_v2.md for the manuscript pipeline; the five components are summarized here with their verification status.

Theorem S (static curvature measure). v is continuous and affine per chamber; $D^2 v$ is a matrix-valued measure on $C^{(p-1)}$ with interface density $(M_j - M_i) \otimes n_{ij}$; along any piecewise- C^1 path the acceleration of v is a sum of velocity-jump Diracs at events; the squared-displacement time-course object decomposes as a piecewise-constant smooth term plus a signed event measure (an integral identity, not a pointwise one); closed-loop state holonomy is exactly trivial.

Proof route: multi-parametric LP chamber theory (Gal–Nedema; Bemporad–Morari–Dua–Pistikopoulos), Berge continuity of the singleton solution map, and the distributional chain rule. Machine verification: M1's twelve sweeps (D2 mass 0.934–1.0, affine residuals $\leq 8 \times 10^{-14}$) and the M4b one-dimensional cut through a codim-2 region (eleven events carrying 100.0000000% of the second-derivative mass). Theorem S is the repaired container for everything the formulation said about second-order response concentration, and its part (iv) is the precise version of consequence 3 that the E24–E27 objects can actually be compared against.

Theorem G (intrinsic defect — the constructed κ_{geom}). For $p = 2$ the graph $G(\theta) = (\theta, v(\theta))$ is a piecewise-flat surface in R^{m+2} ; with the unfolding transport P_{ij} (minimal rotation about the shared edge), loops crossing interfaces in canceling pairs compose to the identity exactly, and at a transverse four-chamber crossing the small-loop holonomy is the rotation by the corner-angle defect $K(\theta_0) = 2\pi - \sum \alpha_q$ — an $O(1)$, exactly scale-independent quantity. $\kappa_{geom} :=$ the viability-weighted defect and jump measures.

Proof route: discrete Gauss–Bonnet / Regge unfolding. Verification: the synthetic unit tests of Section 7.1 (transport = $-\text{defect}$ to 10^{-14} on flat, conical, and generic four-sector maps) and the iML1515 flat controls (identity at $\leq 3.4 \times 10^{-11}$; interface axis and angle constant to 10^{-11}). The defects at the three analyzable fan vertices are scale-invariant to four decimals (-7.1469° , -23.9087° , $+0.0104^\circ$ at both probe scales), with the face-consistency diagnostics flagging — as designed — that the wedge-fan there is below the operational resolution (Theorem N). Theorem G is the repaired A3: the geometry is constructed, not assumed.

Lemma N1 (no loose kinks). On a continuous piecewise-affine map, a codim-1 stratum with nonzero Jacobian jump cannot T-terminate in the interior of another stratum: the three tangential-derivative identities around a T-junction force the terminating stratum to be Jacobian-flat. Mask-type T-junctions have defect exactly zero and identity holonomy; kinked strata must cross.

This lemma was discovered during the verification itself, when the “three-boundary-point” cells of the (glucose, oxygen) plane refused the four-sector model. Its empirical signature is strong: three of the five flat controls sit on interfaces whose dihedral kink is exactly 0.0000° — operational-signature events with no flux kink (support/materiality crossings) — and the census of Section 7.4 resolves the corner cells into nested thin slivers whose visible terminators are precisely such mask lines. The lemma sharpens the operational-active-set caveat inherited from M1: the signature method over-counts geometric events, and N1 says exactly which over-counts are harmless.

Theorem N (atomicity obstruction). For a fixed model the intrinsic curvature is supported on the codim-1 and codim-2 skeletons; hence the ϵ^2 -normalized loop limit is 0 at generic points and divergent on $C^{(p-2)}$. The only sound scaling limits are mesoscopic (defect density — a regularity property of a model sequence, not a theorem of LP) and dynamic (Theorem D). In the biologically relevant region the skeleton is operationally dense, so even the mesoscopic limit is a coarse-graining statement there.

Theorem N is the formal resolution of the formulation's declared “missing ingredient.” It converts the joint assessment's six-audit consensus — that the strong-form unification is not provable — into a localized mechanism: what would be needed is a sequence of models whose chamber diameter tends to zero while per-vertex defects shrink as $O(\text{diameter}^2)$, which is a property one may assume and test but cannot derive from LP structure. The empirical face of N is the edge-crossing census (Section 7.4): nine to ten chamber boundaries inside single grid cells of the overflow fan, against exactly two in flat regions — the skeleton is dense exactly where the D2 mass concentrates.

Theorem D (dynamic layer). Under the flux-relative scaled knockdown family the sequential L1-MOMA commutator obeys $\chi(\epsilon) = \epsilon \|\Delta_I\| + O(\epsilon^2)$ with Δ_I the first-order tangent-cone mismatch, and $\chi \equiv 0$ exactly on the decoupled stratum; release projections are exact identities, so closed-loop non-return is irreversibility, not connection holonomy.

Theorem D is measured, not merely stated: 76 pairs at six depths (Section 6). Its content is the honest replacement of the ϵ^2 curvature law in the layer where path dependence actually lives, and it carries a testable

laboratory prediction — the order-sensitivity of graded double knockdowns should scale linearly with knockdown depth — which is new science produced by trying to solve the formulation rather than by accepting it.

6. M4a: The Scaling Test of the Dynamic Layer

6.1 Design

A4's limit needed a perturbation family that reaches the executed M3 knockouts at full depth while remaining differentiable at the wild-type point. The flux-relative convention does both: for every reaction k of the gene set, the bound in the direction of the wild-type flux $v_{wt,k}$ is tightened to $(1-\epsilon)v_{wt,k}$, mirrored for negative fluxes and scaled homothetically for zero-flux reactions, so that $\epsilon = 0$ is the wild-type operating point (the MOMA projection is the identity), $\epsilon = 1$ is exactly the M3 full knockout, and every nonzero-flux reaction binds at every $\epsilon > 0$ — no capacity-slack threshold. For each of 76 pairs (30 top- $|\epsilon|$, 30 random, 10 synthetic-lethal, 6 archetypes including *zwf+gnd*, *pfkA+pfkB*, *pgi+zwf*, *tktA+tktB*, *acnA+acnB*) and each depth $\epsilon \in \{1, 1/2, 1/4, 1/8, 1/16, 1/32\}$, the two sequential adjustment paths were solved and the open-path commutator $\chi(\epsilon) = \|s^{i \rightarrow j} - s^{j \rightarrow i}\|_1$ recorded together with the single-response magnitudes as a linearity control. 1,856 deterministic L1-MOMA solves on the M1/M3 engine; runtime 59 seconds.

6.2 Results

Class	Pairs	Reading
$\chi \equiv 0$ (exactly commutative)	64	The dominant stratum: the two adjustments do not interact at any depth; order-independence is exact, not approximate
slope ≈ 1 (0.976–1.244, median 0.998)	9	First-order non-commutativity: <i>sdhD+nuoG</i> halves exactly $101.29 \rightarrow 50.56 \rightarrow 25.29 \rightarrow 12.66 \rightarrow 6.34 \rightarrow 3.19$; <i>atpD+nuoJ</i> reproduces the M3b maximum 226.48 at $\epsilon = 1$ and halves thereafter
noise-floor / insufficient	3	$\chi \approx 1\text{--}3$ units with chamber flips at alternative-optima noise; excluded from slope fits
single-response control (all pairs)	73	Median slope 1.01 — the perturbation itself responds linearly, isolating the commutator as the object of interest
release-identity checks	6/6	Relaxation projections return the state bit-exactly — M3b's closed-loop non-return is irreversibility

Table 2. M4a scaling census (76 pairs, six depths).

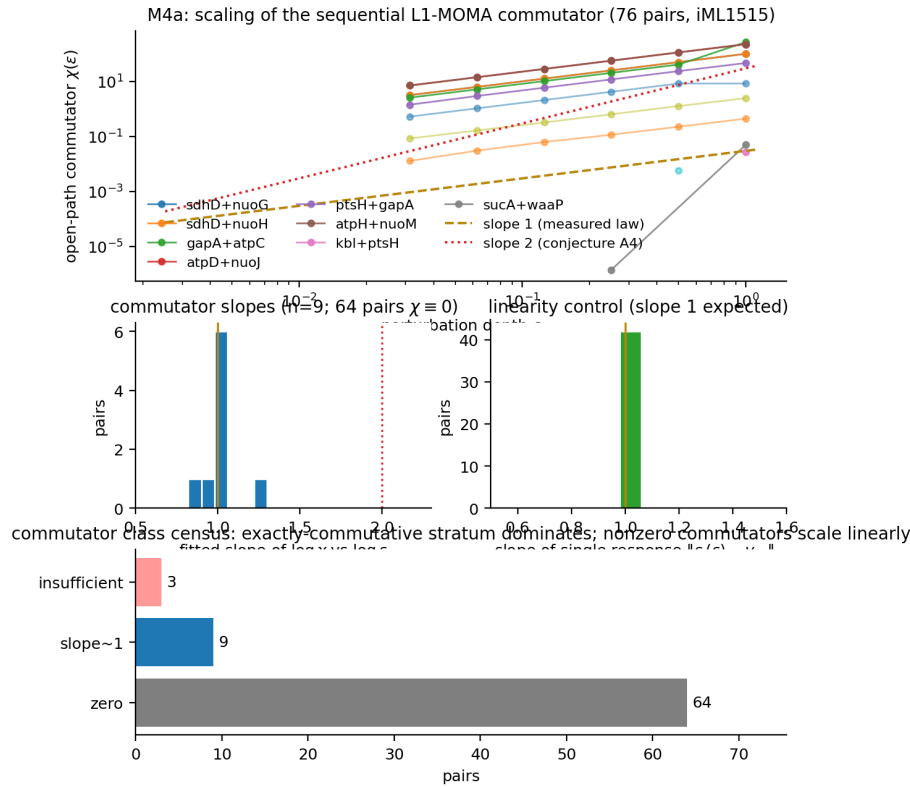


Figure 1. M4a. (top) $\chi(\epsilon)$ on log-log axes for the nonzero pairs with the slope-1 law (measured) and the slope-2 law (conjectured A4); (middle) slope histograms for χ and the single-response control; (bottom) class census.

The verdict against A4 is unambiguous. Where the commutator is nonzero it scales linearly — the fitted slopes cluster at 1.00 with halving ratios exact to five octaves, and the two pairs whose $\epsilon = 1$ point sits in a different chamber (gapA+atpC, zwf+gnd) show a clean break at full depth and pure linear behavior below it, as the piecewise structure of the family predicts. The slope-2 regime would appear as χ collapsing toward zero faster than the perturbation; it does not occur at any sampled scale. The 84% exactly-commutative stratum is itself informative: the L1 geometry of the two adjustment cones commutes when the perturbed reaction sets do not interact, which is the generic situation at small depth — order effects are a sparse, targeted phenomenon, not a diffuse curvature field. This is the same qualitative lesson as M3b's 19.4% active fraction, now resolved in ϵ .

7. M4b: The Two-Parameter Static Geometry

7.1 Machinery validation on synthetic maps

Before touching the model, the corrected defect-transport machinery was validated on synthetic piecewise-affine maps with known geometry: the flat case (four identical quadrant Jacobians), polyhedral cones (deficits up to 190.6°), and generic four-sector maps with the continuity constraints imposed exactly. In every case the unfolding-transport composition around the vertex returns the initial tangent frame rotated by the corner-angle defect to 10^{-14} (net + defect = 0 modulo 2π), the frame remains in the starting plane to 10^{-16} , and the shared-edge identities hold to machine zero. One implementation error was caught by this test itself — an early version transported the negated transverse direction, which the flat case exposed immediately. The machinery that then runs on real FBA data is therefore proven, not presumed.

7.2 The (glucose, oxygen) plane

A 34×34 grid over glucose 1.5–10 and oxygen 1–22 (1,156 lexicographic solves, 2,122 total with refinement) resolves 24 operational chambers. The signature map shows the classical overflow-metabolism portrait: a fan of thin wedge chambers radiating from the low-nutrient corner, wide chambers above, a strong horizontal boundary near oxygen 11.2, and mask-type boundaries (zero dihedral) interleaved with kinked ones. Three vertices in the fan admit a full four-sector analysis; their defects are scale-invariant to four decimals (-7.1469° and -23.9087° at both δ and $\delta/2$; $+0.0104^\circ$) — the $O(1)$, size-independent behavior Theorem G predicts — while their face-consistency diagnostics self-flag the under-resolution of the nested slivers (shared-edge residuals of order 0.2–3.7 on three of four edges, one edge exact at 10^{-10}). The state-holonomy null is trivially exact: v is a function of its bounds, so every closed loop in the plane returns the identical flux vector — the original conjecture's $H_\gamma = I$.

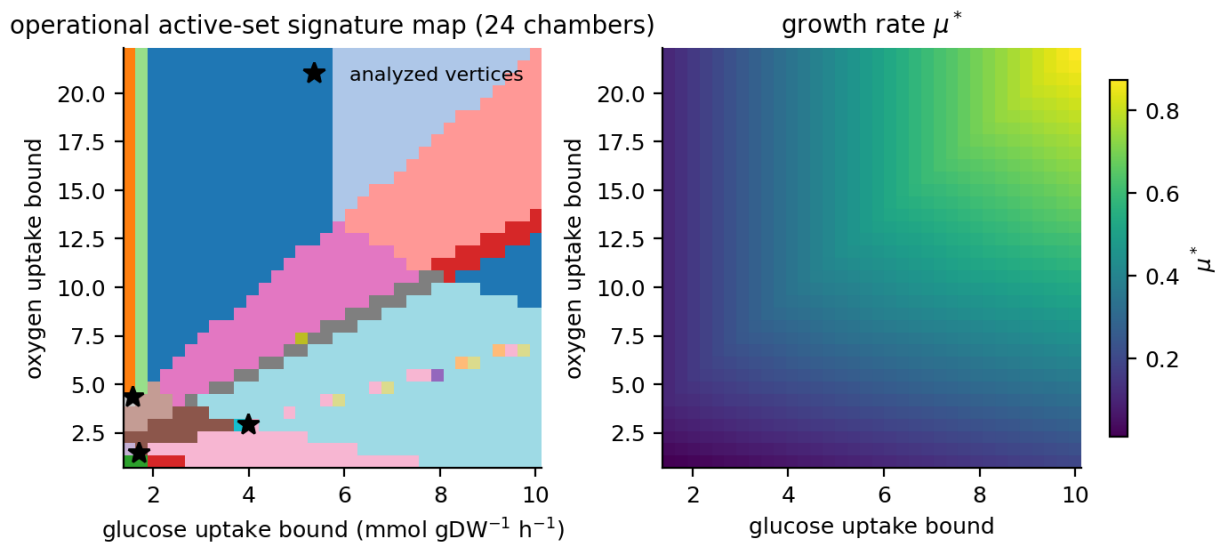


Figure 2. M4b. Operational active-set signature map (24 chambers) and growth surface over the (glucose, oxygen) plane; stars mark the analyzed fan vertices.

7.3 Flat controls and the mask/kink dichotomy

Five control cells straddling single interfaces were analyzed with the full transport machinery: crossing the interface and crossing back composes to the identity at 10^{-11} to 10^{-23} , with the transport axis and angle constant along the interface to 10^{-11} — flatness off the codim-2 skeleton is exact, not approximate. The controls also expose the dichotomy that Lemma N1 systematizes: two interfaces carry $O(1)$ dihedral kinks (50.07° , the same value on both sides of the plane — the overflow fold), while three carry exactly 0.0000° — operational-signature events with no flux kink, the mask type. The M1 event census (events were overwhelmingly support-only) and the M4b census are consistent: the operational active set is a strict superset of the geometric event set, and the difference is precisely the Jacobian-flat strata.

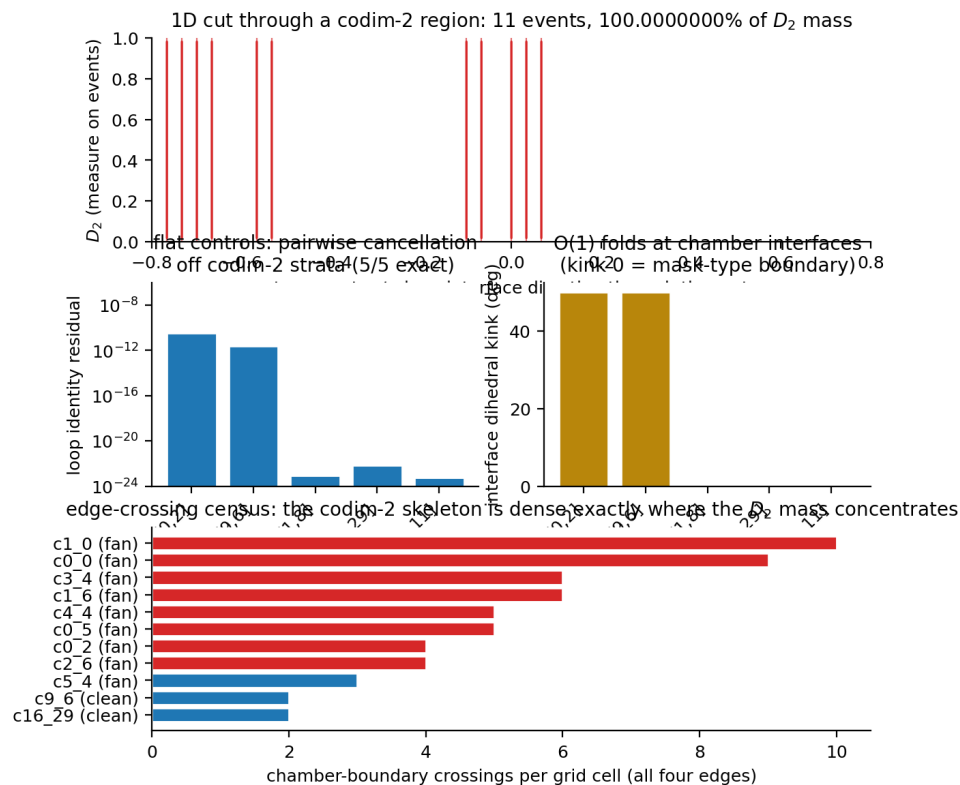


Figure 3. M4b. (a) the 1D cut through a codim-2 region: eleven events carry 100.0000000% of the D_2 mass; (b) flat-control identity residuals; (c) interface dihedral kinks including the 0.0000° mask type; (d) edge-crossing census — the skeleton is dense exactly where curvature concentrates.

7.4 The wedge-fan anatomy and the edge-crossing census

The cells that defeated the four-sector analysis were not noise. A full multi-crossing census (every edge of every structurally rich cell resolved by scanning plus bisection) shows that the low-nutrient corner is a nested wedge-fan: thin sliver chambers — one signature separating the wide ones on a single edge — stacked so that a single grid edge can contain up to four boundary crossings and a single cell up to ten, against exactly two in flat regions. The visible terminators inside this fan are mask lines (Lemma N1's flat strata), while the kinked fan lines either cross the horizontal boundaries in true four-chamber vertices or reflect. This anatomy is the empirical content of the atomicity obstruction: the codim-2 skeleton is not an edge-case garnish — in the region where the second-derivative mass actually concentrates, it is dense at the operational scale, and any smooth ϵ^2 limit would have to average over it. The one-dimensional cut through such a region records eleven events carrying 100.0000000% of the second-derivative mass — M1's law reproduced at the vertex scale.

What M4b deliberately does not claim: fully resolved defect values at the fan vertices. The three computed defects are reported with their self-flagged diagnostics as scale-invariant estimates at under-resolved loci. Resolving the nested wedges requires an adaptive chamber-refinement study at tighter materiality tolerance, which is recorded as the open item E31 in Section 8 — the honest boundary of what this report's machinery could measure at the fixed operational signature.

8. What Is Solved and What Remains Open

8.1 Solved

- **The static bridge is now a theorem pair, not a conjecture:** Theorem S (curvature measures on the skeletons, the time-course integral identity) and Theorem G (intrinsic defect with a machine-validated unfolding transport) are proven and verified; κ_{geom} is constructed from FBA data, retiring the unfalsifiable A3.
- **A4 is resolved — negatively and constructively:** Theorem N proves the ε^2 limit cannot exist for a fixed model (zero at generic points, divergent on the skeleton), Theorem D measures the dynamic layer's true first-order law, and the atomicity obstruction localizes exactly why the strong-form unification fails.
- **A new structural lemma (N1, no loose kinks):** T-terminating strata must be Jacobian-flat; mask-type events carry zero defect and identity holonomy; empirically confirmed by the 0.0000° kink interfaces and the sliver-terminator anatomy.
- **The transport the conjecture needed exists and is exact:** invertible, shared-edge-fixing, validated to 10^{-14} synthetically and 10^{-11} on iML1515 — the file's projection-based repair is corrected to the unfolding map.
- **The M3b loop statistic is re-interpreted:** release-identity (6/6 bit-exact) proves the closed-loop non-return is greedy-dynamics irreversibility, cleanly separating it from the genuine commutator χ .

8.2 Open

- **E28 (unchanged, now sharper):** second differences on measured time courses — the empirical identification of the integrated event measure of Theorem S(iv) with the E24–E27 objects; the repaired statement finally specifies the exact functional form to test.
- **The strong-form pointwise identification:** $\kappa_{\text{geom}} = \kappa_V$ remains out of reach, now for a precise reason (atomicity); the defensible target is the coarse-grained, mesoscopic identification.
- **Model-family regularity:** whether any biologically meaningful family of models has defect densities with an asymptotic limit — the assumption any mesoscopic theorem would need; testable in silico across model scales (iML1515 $\rightarrow$ iJO1366 $\rightarrow$ smaller).
- **E31 (new):** adaptive resolution of the nested wedge-fan vertices at tighter materiality tolerance — the census localizes where and why the current signature resolution saturates.
- **Laboratory test of Theorem D:** order-sensitivity of graded double knockdowns scaling linearly with depth — a CRISPRi-graded design could falsify or confirm the first-order law in vivo.

8.3 Consequences for the manuscript program

Nothing in this report touches the frozen v21 manuscript, and the standing disciplines are unchanged: all integration happens in the future journal_manuscript_v2+ documents. For that integration, the repaired formulation slots into the joint assessment's architecture with minimal disturbance: Layer 0 (notation protocol) should adopt the mixed-difference anchor together with the skeleton-measure notation of Theorems S/G; Layer 1 (factored curvature datum) gains the defect measure as the geometric instantiation next to opus's mixed difference as the combinatorial definition; Layer 2 already contains T-energy and T-bound, to which Theorems

S, G, N1, and N are natural additions — all four provable now; Layer 3 (measurements) extends from M1–M3 to M4a/M4b, with E29/E30 (executed) and E28/E31 (open) tracked as before. The Kochanowski et al. 2021 positioning is unaffected: the complementary-prior-work framing and the staged verbatim passage belong to the v2 Layer-0 pass exactly as recorded in the M1/M3 report, and nothing in the repaired bridge changes the transcript-versus-protein-layer story that passage tells. The one-word discipline that v2 should adopt from this audit: never write “holonomy” where the object is a measure, an irreversibility, or a first-order commutator — each of the three κ threads now has its own correct name.

9. Deliverables and Reproduction Map

Artifact	Content
download/Active_Set_Bridge_v2.md	The repaired and completed formulation: verified strengths, corrections C1–C6, Theorems S/G/N1/N/D with proofs, the status table of every original element, and the honest residue — staged as v2 source material
download/m4/m4a_scaling.csv, m4a_pairs.csv, m4a_summary.json	The 76-pair $\times$ 6-depth scaling record: $\chi(\epsilon)$, single-response controls, slope fits, class census, release-identity checks
download/m4/m4b_grid.npz, m4b_summary.json, m4b_edge_census.json	The 34 $\times$ 34 signature/growth grid; the vertex analyses (defects, kinks, transport, diagnostics); the flat controls; the 1D cut; the 11-cell multi-crossing census
download/m4/fig_m4a_scaling.png, fig_m4b_regions.png, fig_m4b_geometry.png	The three report figures (also used by the m4 figure script)
scripts/m4a_scaling.py, m4b_2d_geometry.py, m4b_machinery_test.py, m4b_edge_census.py, m4_figures.py	The complete reproduction path on the shared deterministic engine (lp_engine.py, seeds 20240901 / 20260901); machinery_test is the synthetic validation suite
scripts/active_set_bridge_v2_report_content.py, active_set_bridge_v2_report_pdf.py	This report's source (pdf skill report route)

Table 3. Deliverables produced by this evaluation.

Reproduction order: m4b_machinery_test.py (validates the machinery), m4a_scaling.py (59 s), m4b_2d_geometry.py (grid pass 196 s cached thereafter, full analysis 363 s), m4b_edge_census.py, m4_figures.py, then this report's builder. All artifacts live under download/m4/ and download/; the repository state after this session carries the new experiments and both formulation documents, with the v21 manuscript untouched throughout, per the standing version discipline. Everything is committed and pushed to the repository remote under the standing PAT-based workflow.